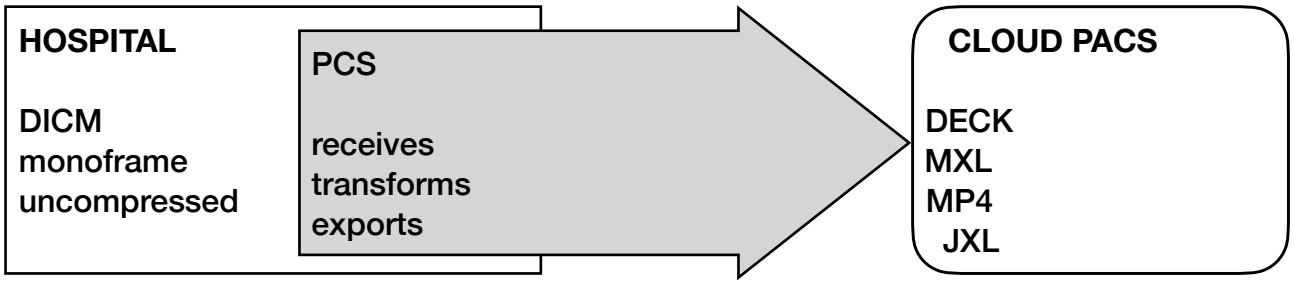
PCS DICM processing

Integration



Objectives

- Preprocess DICM hospital side, simplifying the task of the PACS in the CLOUD. This design escalates naturally with the number of hospitals served and lowers processing cost on the cloud side.
- Separate data and metadata
- use series level web optimized compressed formats for data
- transform CT, MR, PT series into converted enhanced ones (one multiframes instead of many monoframe)
- use DECK multilevel (study, series, instance, private) for metadata. The pacs receives key-value metadata already formated for inclusion into a sqlite database.

3 asynchronous processes

(1) reception, parsing, pixels in tiff, htj2k or jxl
(2) consolidation of series frames into mp4 av1, mxf htj2k or jxl + eventual consolidation of CT, MR, PT into converted enhanced
(3) forwarding to cloud with some kind of rsinc function

(1) reception, parsing, pixels in tiff, htj2k or jxl

The reception is performed by dcmtk storescp in explicit little endian, which finalizes its process forking another one executing a script.
This script processes this raw DICM instance and in case of success, moves the original out of the inbox.
The script has one file as input, parses it and as a result creates a file system as output. Parsing is transforming a language into another, easier to process. The modularization of the original file into a file system belongs to the parsing process.
Another part of it rewrite DICM syntax into DECK flat key value one.
Finally, the parsing transforms the raw pixel data into universal picture formats, which contain minimal strictly image presentation metadata.

In the case of direct X-ray and similar where each image is independent, there remain the possibility that the all the different instances where kept in a unique series. In order to simplify comparative hanging protocols, we create new series one for each image starting with the second one of the series.

In the case of original multiframes, each frame is transformed into an individual picture format (tiff for further av1 consolidation or htj2k for further mxf consolidation). Regarding matadata, the converted enhanced version gathers instance specific attributes into sequences. To build the

sequence, specific accumulative files are created to append the corresponding attribute of each instance.

In the case of related instances to be transformed into a converted enhanced CT, MR, PT, the images are gathered inside a special directory at series level.

(2) consolidation of series frames into mp4 av1 or mxf htj2k or jxl + eventual consolidation of CT, MR, PT into converted enhanced

Prerequisite: all the instances of the series have been successfully parsed (process 1). In the case of a series of monoframes, this prerequisite is difficult to validate. A good approximation includes:

- existing image numbers start at 1 and increase arithmetically, without missing number within the series
- an instance of another series of the study, or of another study has been received.
- there has been no new instance received for x seconds